Solution Requirements

Date	29 September 2026
Team ID	SWTID-2026-8067
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The solution requirements define what the ServiceNow ACL solution must do and the quality criteria used to verify secure and reliable access to Institution Details records.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users must sign in to the ServiceNow instance with an identifiable user account before accessing the protected table.	High
2	Authorization levels	bb1 controls Read access; bb2 controls Create; bb3 controls Write; bb4 controls Delete. Administrators retain full access.	High
3	External interfaces	No external API is required for this lab. The solution uses native ServiceNow users, roles, ACLs and table records.	Low
4	Transactions processing	Read, Create, Write and Delete requests are evaluated by ServiceNow ACLs before the operation is permitted.	High
5	Reporting	The solution should allow administrators to inspect the Institution Details list and verify which records are accessible during testing.	Medium
6	Business rules, etc.	Read access is restricted to records where Branch = EEE for authorized bb1 users; administrator access is allowed. CRUD operations use separate roles.	High
7	Compliance to laws or regulations	The lab should follow the organization's applicable data-protection and access-control practices for student/institution information.	Medium
8	Other	Impersonation must be used to verify behavior for authorized, unauthorized and administrator users.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	ACL evaluation should occur as part of normal ServiceNow access checking without unnecessary custom processing.	Access decision completes during the normal request/list/form operation.
2	Scalability	The solution should use native ServiceNow ACL and role mechanisms rather than client-side filtering.	Security remains server-side and applies consistently to protected records.
3	Security & Data Privacy	Unauthorized users must not gain Read, Create, Write or Delete access through the protected ACLs.	Users without required roles are denied; bb1 users see only permitted EEE records; admins retain full access.
4	Reliability & Availability	Access rules should remain active and consistently enforced across supported ServiceNow interfaces.	Same ACL rules are evaluated for list/form/record operations.
5	Usability & Accessibility	Authorized users should see only the actions and records allowed by their roles, while administrators can manage the configuration.	New/Edit/Delete availability matches the evaluated ACL permissions.
6	Maintainability	Roles and operations are separated into bb1, bb2, bb3 and bb4 so individual permissions can be changed independently.	Each ACL has a clear operation and required role.